

Primary/secondary node IP addresses		
Parameters / Node	rproxy1.unix-clinic.net	rproxy2.unix-clinic.net
eth0 (192.168.1.x is Private Subnet for Heartbeat)	192.168.1.1	192.168.1.2
Eth1 (administrative address)	172.202.2.1	172.202.2.2
Service address 10.8.0.1	Primary node	Secondary node

Table 1

The architecture: active-passive HA cluster

The two primary modes of an HA cluster are:

- **Active-passive:** In Active-Passive HA clusters, the primary node is active and serves the request. In case it fails, then the services are transferred to the secondary node, either through automatic or manual failover.
- **Active-active:** In active-active HA clusters, both the nodes remain active all the time and serve their respective requests. In case one of the nodes goes down, the services running on that node are failed-over to the other node in the cluster. This way an active-active HA cluster is used when you have multiple services under the high-availability requirement.

The service being served by the HA cluster depends on the IP address, so we first need to distinguish between the 'administrative address' and the 'service address'. Each interface on the cluster nodes should have an administrative address and can optionally have one or more service addresses, depending on the cluster configuration (active-active or active-passive) and state (active or standby).

An administrative address is one that is in control of the operating system and is brought up and down with the OS. A service address is one that is under the control of the Heartbeat software, which then controls its allocation to one of the cluster nodes. The node where this service address should reside, by default, is known as the primary or the active node, and the other node in the cluster is known as a secondary, failover or passive node.

In failover clustering, when a failover happens, the secondary node takes over the service address and becomes active. This is how we will be configuring our cluster.

In our case, the service being offered by the cluster, a reverse-proxy server, depends on the IP address. So we need to take care of the following points:

- Make sure that the nginx server is not started automatically on any node, but is under the control of Heartbeat.
- The functioning of nginx will be dependent on the availability of the service IP address and hence, in case of a failover, we need to make sure that nginx is started after the service IP address has been taken over by the secondary node.

Table 1 can be referred to for setting up the networking. For a test environment, a flat network would do.

Installation and configuration of Heartbeat

On Debian-based systems, Heartbeat can be installed as follows:

```
# apt-get install heartbeat-2
```

On CentOS 5.2, after subscribing to the 'extras' repository, execute the following command:

```
# yum install heartbeat
```

Typically, in an active-passive HA environment, the nodes of the cluster have an identical set-up, so unless otherwise noted, all the configurations have to be done identically on both the nodes. It is not mandatory that the nodes are of identical hardware configuration, but this is recommended for a production environment. Also, having the same OS on both the nodes will help from the maintenance and troubleshooting point of view.

In the high-availability environment, all nodes should be able to see one another, irrespective of the availability of the DNS; so we need to make sure that there are relevant entries in the `/etc/hosts` file on each node. My host's file for this set-up looks like what's shown below:

```
# cat /etc/hosts
.....
192.168.1.1 rproxy1
192.168.1.2 rproxy2
```

Now you should be able to *ping* both nodes from each other. This will make sure that the Heartbeats from both nodes will see each other, irrespective of the DNS. You can build as many redundancies as you want; e.g., you can have bonding for the interface that's responsible for sending Heartbeat, you can have trunking on the switch ports, and so on. Whatever you do, just make sure it's not an overkill and the set-up is not unnecessarily expensive.

The `ha.cf` file

The main configuration file for Heartbeat is `ha.cf`, which lists the nodes of the cluster, communications topology and all the features that are enabled. The order of directives in `ha.cf` matters, so make sure you take note of that.

The minimum `ha.cf` file will look like the following, on both the nodes:

```
# cat /etc/ha.d/ha.cf
use_logd on
keepalive 2
deadtime 30
warntime 10
initdead 120
udpport 6909
ucast eth0 192.168.1.1
ucast eth0 192.168.1.2
# bcast eth0
```